UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Particular Integral, Type 2: Sine and Cosine Right-Hand Side

Complete study pack for class 004: the rule $\frac{1}{f(D^2)}\sin(ax+b) = \frac{\sin(ax+b)}{f(-a^2)}$, rationalising odd powers of D, the cases of failure $f(-a^2)=0$, and right-hand sides that reduce to Type 2 (products $\sin ax\cos bx$, $\sin^2 ax$, $\cos^2 ax$, $\sin^3 ax$) combined by superposition

Compiled from the PESU Unit 2 class 004 slide deck, its two AV-summary transcripts (the Type 2 lecture and the newer Type 1 and Type 2 problem lecture), the Unit 1 & 2 class-work sheet (problems 13, 14, 18, 21, 22) and eight past ESA papers. Generated 16 October 2026.

· How to use this pack

Topic 1 is the Type 2 rule: replace D^2 (never D) by $-a^2$, then rationalise any odd power of D that is left (Figure F1.1). Topic 2 is the case of failure, $f(-a^2) = 0$, which happens exactly when $D^2 + a^2$ is a factor of $f(D)$, that is when $\pm ai$ are roots of the auxiliary equation (Figures F2.1, G2.1, G2.2); it also solves the ESA question $(D-2)^2 y = 8(e^{2x} + \sin 2x)$, asked three times. Topic 3 reduces products and powers of sines and cosines to Type 2 and adds the PIs of the separate terms (Figure F3.1, table C3.1). The complementary function rules are in the class 002 pack and the Type 1 rule (e^{ax}, constants) in the class 003 pack; both are used here without derivation. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9	Unit 2 deck 004 (“Particular integral of Type 2”), slide 9 (rendered .pptx)
T004 00:10:32	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 2” (24.7 min), segment start
T004b 00:08:13	Unit 2 AV-summary transcript “Particular integral Type 1 and Type 2” (12.6 min, newer recording); its Type 1 part is tagged T004a in the class 003 pack
CW U2 P13	Unit 2 class-work problem 13
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

Contents

1. 1 · The Type 2 rule: replace D^2 by $-a^2$
2. 2 · Case of failure: $f(-a^2) = 0$
3. 3 · Products, powers and sums: reducing to Type 2
4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

1 The Type 2 rule: replace D^2 by $-a^2$

When $X = \sin(ax+b)$ or $\cos(ax+b)$, the PI $\frac{1}{f(D)}X$ is found by replacing every D^2 in $f(D)$ by $-a^2$ (minus a squared), provided the result is not zero. Odd powers are first written through D^2 ($D^3 = D^2 \cdot D$), so a single D can remain; it is removed by multiplying above and below by the conjugate (rationalising), which turns the denominator into a function of D^2 again. The PI always has the same frequency a as X, only a new amplitude and phase.

Key terms and definitions

Type 2 right-hand side. $X = \sin(ax+b)$ or $X = \cos(ax+b)$, with constants a and b. S004·3 · S004·4 · T004 00:00:00

$f(-a^2)$. The expression obtained from $f(D)$ by replacing only the D^2 terms by $-a^2$ (minus a squared). D^3 is written $D^2 \cdot D$ and D^4 as $(D^2)^2$ first. T004 00:01:15 · T004 00:01:53 · T004b 00:06:32

Rationalising (conjugate). If a single D remains, as in $\frac{1}{p+qD}$, multiply numerator and denominator by $p-qD$ to get $\frac{p-qD}{p^2-q^2D^2}$; then replace D^2 by $-a^2$ again and let $p-qD$ act on X (differentiate). S004·9 · T004 00:11:48 · T004b 00:06:59

> *"Type (ii); When $X = \sin(ax+b)$ or $\cos(ax+b)$: $\frac{\sin(ax+b)}{f(D^2)} = \frac{\sin(ax+b)}{f(-a^2)}$ provided $f(-a^2) \neq 0$; $\frac{\cos(ax+b)}{f(D^2)} = \frac{\cos(ax+b)}{f(-a^2)}$ provided $f(-a^2) \neq 0$."* S004·4

Formulas

E1.1 · Type 2 rule ESA ★★★

$$\frac{1}{f(D^2)}\sin(ax+b) = \frac{\sin(ax+b)}{f(-a^2)}$$
$$\frac{1}{f(D^2)}\cos(ax+b) = \frac{\cos(ax+b)}{f(-a^2)}$$

gives	The PI of a sine or cosine.
applies	$f(-a^2) \neq 0$, i.e. $\pm ai$ not roots of the AE.
note	Replace D^2 by $-a^2$, not by $(-a)^2$; b plays no part.

S004·4 · S004·15 · T004 00:00:58 · T004b 00:06:32

E1.2 · Writing odd powers through D^2 ESA ★★★

$$D^3 = D^2 \cdot D \to -a^2 D, \qquad D^4 = (D^2)^2 \to a^4, \qquad D^5 = (D^2)^2 D \to a^4 D$$

gives	A denominator of the form $p+qD$.
applies	$f(D)$ has odd powers of D.
note	Lecture example: $D + D^2 + D^3 \to D - a^2 - a^2 D$.

T004 00:01:42 · T004 00:11:10 · S004·9

E1.3 · Rationalising a linear denominator ESA ★★★

$$\frac{1}{p+qD}X = \frac{(p-qD)}{p^2-q^2D^2}X = \frac{(p-qD)X}{p^2+q^2a^2}$$

GIVES A PI with no D left in the denominator.

APPLIES After E1.2 a single D remains; $X = \sin(ax+b)$ or $\cos(ax+b)$.

NOTE $p-qD$ then acts on X: $D\sin(ax+b) = a\cos(ax+b)$, $D\cos(ax+b) = -a\sin(ax+b)$.

S004·9 · S004·17 · T004 00:12:01 · T004b 00:11:14

Graphs and diagrams

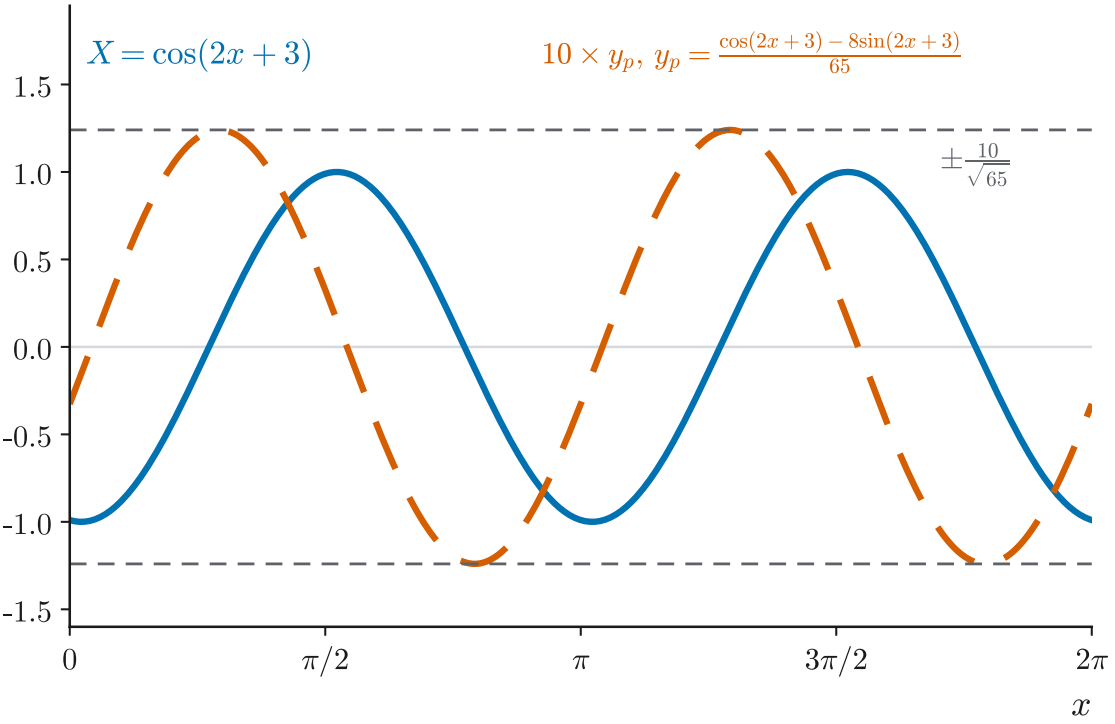
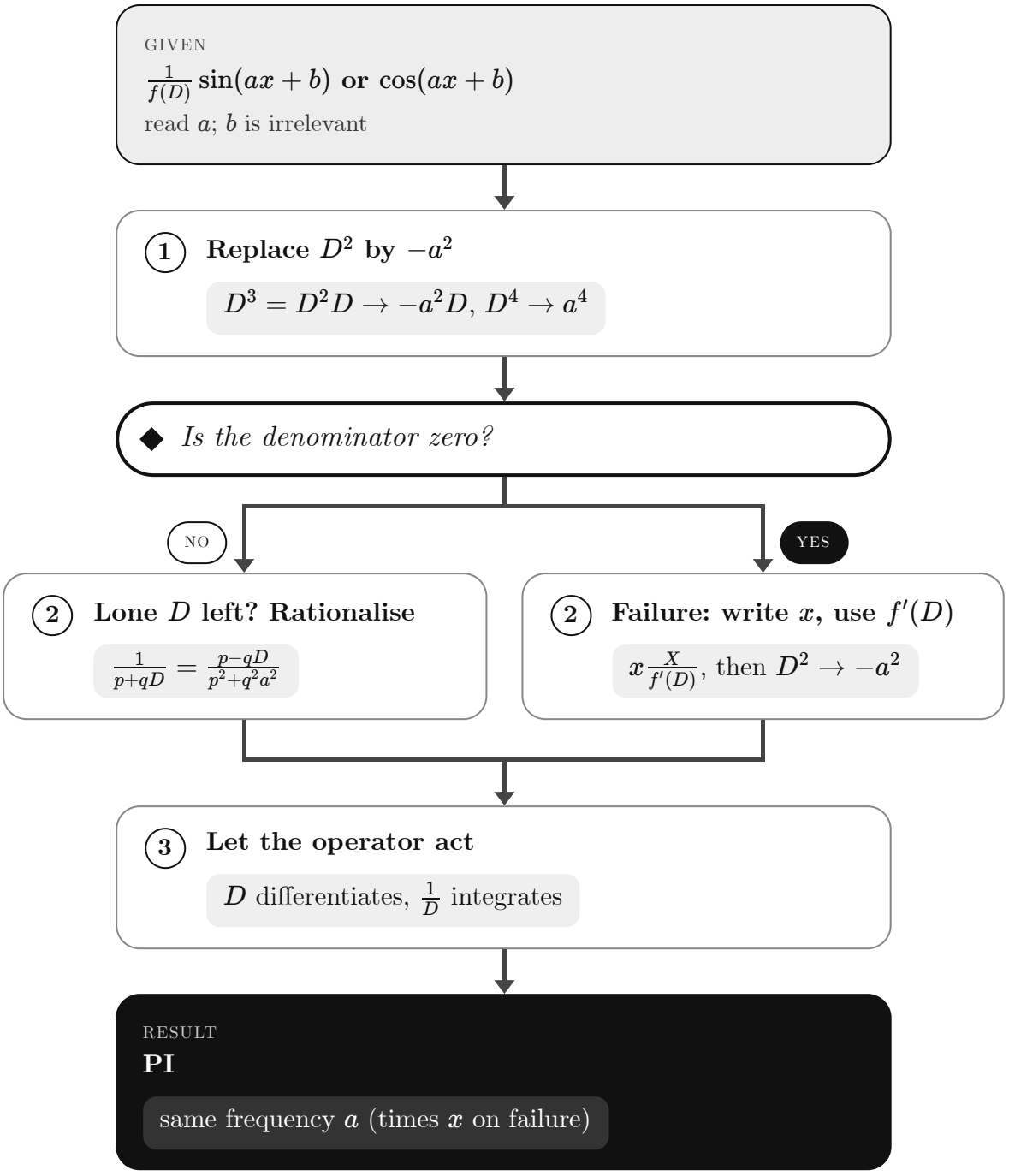


Figure G1.1. Same frequency, new amplitude and phase: the PI of $(D^3+1)y = \cos(2x+3)$. ESA ★★

ADDED FOR CLARITY S004·8 · S004·9

How to read it. The forcing $\cos(2x+3)$ (blue) and ten times its PI $\frac{\cos(2x+3)-8\sin(2x+3)}{65}$ (vermilion dashed). Both have period π; the PI is a single sinusoid of amplitude $\frac{\sqrt{1+64}}{65} = \frac{1}{\sqrt{65}}$ (grey lines, scaled by 10) shifted in phase. This is what the rationalised answer $\cos - 8\sin$ means.

Flowcharts



Remark. The CF warns you: if $\pm ai$ are AE roots, the answer is "yes".

Exam trap. $-a^2$, not $(-a)^2$.

Figure F1.1. The Type 2 fork: does $f(-a^2)$ vanish? ESA ★★★

S004·4 · S004·5 · T004 00:01:15 · T004 00:11:48 · T004b 00:07:32

Type 1 and Type 2 side by side

	Type 1: $X = e^{ax}$	**Type 2: $X = \sin(ax+b)$, $\cos(ax+b)$**
Replace	D by a	D^2 by $-a^2$ (odd D kept, then rationalised)
PI (no failure)	$\frac{e^{ax}}{f(a)}$	$\frac{X}{f(-a^2)}$
Fails when	a is an AE root	$\pm ai$ are AE roots ($D^2 + a^2$ divides f)
First failure	$x\frac{e^{ax}}{f'(a)}$	$x\frac{X}{f'(-a^2)}$
Second failure	$x^2\frac{e^{ax}}{f''(a)}$	$x^2\frac{X}{f''(-a^2)}$

S004·4 · S004·5 · T004b 00:00:41 · T004b 00:06:12

Important points and why

1. Replace only D^2, never D. Write $D^3 = D^2 \cdot D$ and replace the D^2 part. ESA ★★★ T004 00:01:15 · T004 00:01:53 · T004b 00:06:32
2. "It is not minus 2 the whole square. It is minus into 2 square, which is minus 4." ESA ★★★ T004 00:11:33 · T004b 00:06:32
3. If a D remains after the replacement, rationalise with the conjugate so that D^2 appears, then replace again. ESA ★★★ S004·9 · T004 00:11:48 · T004b 00:06:59
4. $\frac{1}{1+4D}$, $(1+4D)$ are operators: the function always stays to the right of the operator. ESA ★★ T004 00:12:15
5. After rationalising, the numerator operator $(p - qD)$ differentiates X: $(1+4D)\cos(2x+3) = \cos(2x+3) - 8\sin(2x+3)$. ESA ★★★ S004·9 · T004 00:13:23
6. The PI has the same frequency as X; a sine input can give a mixture of sine and cosine (a phase shift), Figure G1.1. ESA ★★ ADDED FOR CLARITY S004·9

Why may D^2 be replaced by $-a^2$?
$D^2 \sin(ax+b) = -a^2 \sin(ax+b)$, and the same for cosine, so on these functions D^2 acts as multiplication by $-a^2$. Hence $f(D^2)\frac{\sin(ax+b)}{f(-a^2)} = \sin(ax+b)$. A single D turns sine into cosine, so it cannot be replaced by a number.

ADDED FOR CLARITY S004·4

Why multiply by the conjugate?
$(p+qD)(p-qD) = p^2 - q^2D^2$ contains only D^2, which can be replaced by $-a^2$, a number. The conjugate moves to the numerator, where it is just differentiation.

S004·9 · T004 00:11:48 · T004b 00:11:29

Exam trap. $-a^2$ is not $(-a)^2$. For $a = 2$, $D^2 \to -4$, so $D^3 + 1 \to 1 - 4D$ (slide 004·9). T004 00:11:33 · S004·9

Exam trap. Do not replace a lone D by a or by ia: rationalise it (E1.3). ADDED FOR CLARITY T004b 00:06:59

Worked examples

W1.1 · $(D^3+1)y=\cos(2x+3)$ (slides 004·8–9) ESA ★★★

GIVEN Solve $(D^3+1)y=\cos(2x+3)$.

STEP 1 AE $m^3+1=0$, $(m+1)(m^2-m+1)=0$: $m=-1,\ \frac{1}{2}\pm\frac{\sqrt{3}}{2}i$. CF $=c_1e^{-x}+e^{x/2}\big(c_2\cos\frac{\sqrt{3}}{2}x+c_3\sin\frac{\sqrt{3}}{2}x\big)$.

STEP 2 PI $=\frac{\cos(2x+3)}{D\cdot D^2+1}$; $D^2\to-4$: $\frac{\cos(2x+3)}{1-4D}$.

STEP 3 Rationalise: $\frac{(1+4D)\cos(2x+3)}{1-16D^2}=\frac{(1+4D)\cos(2x+3)}{1+64}$.

STEP 4 $(1+4D)\cos(2x+3)=\cos(2x+3)-8\sin(2x+3)$.

ANSWER $\boxed{y=c_1e^{-x}+e^{x/2}\big(c_2\cos\frac{\sqrt{3}}{2}x+c_3\sin\frac{\sqrt{3}}{2}x\big)+\frac{1}{65}\big[\cos(2x+3)-8\sin(2x+3)\big]}$

CHECK sympy dsolve and checkodesol agree; the slide stops at $y=y_c+y_p$. Figure G1.1.

S004·8 · S004·9 · T004 00:07:01 · T004 00:13:58

W1.2 · $y'''-y''+4y'-4y=\sin 3x$ (CW P13) ESA ★★

GIVEN Solve $y'''-y''+4y'-4y=\sin 3x$, i.e. $(D^3-D^2+4D-4)y=\sin 3x$.

STEP 1 AE $(m-1)(m^2+4)=0$: $m=1,\ \pm 2i$. CF $=c_1e^{x}+c_2\cos 2x+c_3\sin 2x$.

STEP 2 $D^2\to-9$: $-9D+9+4D-4=5-5D$, so PI $=\frac{\sin 3x}{5(1-D)}$ ($3i$ is not a root: no failure).

STEP 3 Rationalise: $\frac{(1+D)\sin 3x}{5(1-D^2)}=\frac{(1+D)\sin 3x}{5\cdot 10}$.

STEP 4 $(1+D)\sin 3x=\sin 3x+3\cos 3x$.

ANSWER $\boxed{y=c_1e^{x}+c_2\cos 2x+c_3\sin 2x+\frac{3\cos 3x+\sin 3x}{50}}$

CHECK CW answer agrees; sympy dsolve and checkodesol agree.

CW U2 P13

W1.3 · Type 2 term of $(D^2-5D+6)y = e^{4x}+\sin x + x^2$ (ESA) ESA ★★

GIVEN Find $\frac{1}{D^2-5D+6}\sin x$ (the full question is solved in the class 005 pack).

STEP 1 $D^2 \to -1$: $-1-5D+6 = 5-5D$, so $\frac{\sin x}{5(1-D)}$.

STEP 2 Rationalise: $\frac{(1+D)\sin x}{5(1-D^2)} = \frac{(1+D)\sin x}{10}$.

ANSWER $$\boxed{\frac{1}{D^2-5D+6}\sin x = \frac{\sin x+\cos x}{10}}$$

CHECK Substituting back gives $\sin x$ (sympy).

ESA Jul-23 3b

W1.4 · The lecture's $D+D^2+D^3$ with $X=\sin 2x$ ESA ★

GIVEN Find $\frac{1}{D^3+D^2+D}\sin 2x$ (the lecture names this $f(D)$ but works no example).

STEP 1 $D^3 = D^2\cdot D \to -4D$, $D^2 \to -4$: $-4D-4+D = -(4+3D)$.

STEP 2 Rationalise: $-\frac{(4-3D)\sin 2x}{16-9D^2} = -\frac{(4-3D)\sin 2x}{52}$.

STEP 3 $(4-3D)\sin 2x = 4\sin 2x - 6\cos 2x$.

ANSWER $$\boxed{\frac{1}{D^3+D^2+D}\sin 2x = \frac{3\cos 2x - 2\sin 2x}{26}}$$

CHECK Substituting back gives $\sin 2x$ (sympy).

ADDED FOR CLARITY T004 00:01:42

Past-paper questions

ESA Jul 2023 · 7 marks · UE22MA141A Q3.b · solved in W1.3

Solve $(D^2-5D+6)y = e^{4x}+\sin x + x^2$. Type 2 part: $\frac{\sin x+\cos x}{10}$. Full solution: class 005 pack.

SHORT-NOTE RECAP

Type 2: $X=\sin(ax+b)$ or $\cos(ax+b)$. PI $=\frac{X}{f(-a^2)}$: replace every D^2 by $-a^2$ (minus a squared), $D^3 = D^2D \to -a^2D$. If a lone D remains, $\frac{1}{p+qD} = \frac{p-qD}{p^2-q^2D^2}$, replace D^2 again, and let $p-qD$ differentiate X. Example: $\frac{\cos(2x+3)}{D^3+1} = \frac{\cos(2x+3)-8\sin(2x+3)}{65}$; CW P13: $\frac{\sin 3x}{D^3-D^2+4D-4} = \frac{3\cos 3x+\sin 3x}{50}$.

2 Case of failure: $f(-a^2) = 0$

If replacing D^2 by $-a^2$ makes the denominator zero, $D^2 + a^2$ is a factor of $f(D)$: $\pm ai$ are roots of the auxiliary equation and $\sin ax$, $\cos ax$ are already in the CF. Then multiply by x and differentiate $f(D)$ with respect to D: $\frac{\sin(ax+b)}{f(D^2)} = x\frac{\sin(ax+b)}{f'(-a^2)}$, and again with x^2 and f'' if f' also fails. The PI then grows like $x\sin$ or $x\cos$ (resonance). The ESA question $(D-2)^2 y = 8(e^{2x} + \sin 2x)$ combines a double Type 1 failure with a Type 2 term.

Key terms and definitions

Case of failure. $f(-a^2) = 0$; it happens when $D^2 + a^2$ is a factor of $f(D)$. S004·5 · T004 00:02:42

$f'(D)$. The derivative of $f(D)$ with respect to D, as if D were a variable: $(D^2+3)' = 2D$. T004 00:03:26 · S004·16

Resonance. Forcing at a frequency already present in the CF; the PI carries an extra factor x and its amplitude grows without bound. (added for clarity) S004·5

> *"Case of failure; If $f(-a^2) = 0$, then $\frac{\sin(ax+b)}{f(D^2)} = x.\frac{\sin(ax+b)}{f'(-a^2)}$ provided $f'(-a^2) \neq 0$; $\frac{\cos(ax+b)}{f(D^2)} = x.\frac{\cos(ax+b)}{f'(-a^2)}$ provided $f'(-a^2) \neq 0$."* S004·5

> *"If $f'(-a^2) = 0$, then $\frac{\sin(ax+b)}{f(D^2)} = x^2.\frac{\sin(ax+b)}{f''(-a^2)}$ provided $f''(-a^2) \neq 0$ (and the same for $\cos(ax+b)$); "similarly, it proceeds on when $f''(a) = 0$....."."* S004·6 · S004·15

Formulas

E2.1 · First case of failure ESA ★★★

$$f(-a^2) = 0: \quad \frac{1}{f(D^2)}\sin(ax+b) = x\,\frac{\sin(ax+b)}{f'(-a^2)} \quad \text{(same for } \cos\text{)}$$

gives The PI when the frequency is in the CF.

applies $f(-a^2) = 0$, $f'(-a^2) \neq 0$.

note f' = derivative w.r.t. D; it usually leaves a D (e.g. $2D$), which is then integration $\frac{1}{D}$.

S004·5 · S004·15 · S004·16 · T004 00:03:07 · T004b 00:07:32

E2.2 · Standard results for $D^2 + a^2$ ESA ★★★

$$\frac{1}{D^2+a^2}\sin ax = -\frac{x\cos ax}{2a}, \qquad \frac{1}{D^2+a^2}\cos ax = \frac{x\sin ax}{2a}$$

gives The failure PI in one line.

applies $f(D) = D^2 + a^2$ (or this factor, with the rest applied first).

note From E2.1: $\frac{x}{2D}\sin ax = \frac{x}{2}\int \sin ax\,dx$.

(added for clarity) S004·16 · T004 00:23:17

E2.3 · Second case of failure ESA ★

$$f'(-a^2)=0: \quad \frac{1}{f(D^2)}\sin(ax+b)=x^2\,\frac{\sin(ax+b)}{f''(-a^2)}$$

GIVES The PI when D^2+a^2 is a repeated factor.

APPLIES $f(-a^2)=f'(-a^2)=0$, $f''(-a^2)\neq 0$.

NOTE And so on: x^3 and f''' if f'' also fails.

S004·6 · S004·15 · T004 00:04:23 · T004 00:05:06

Graphs and diagrams

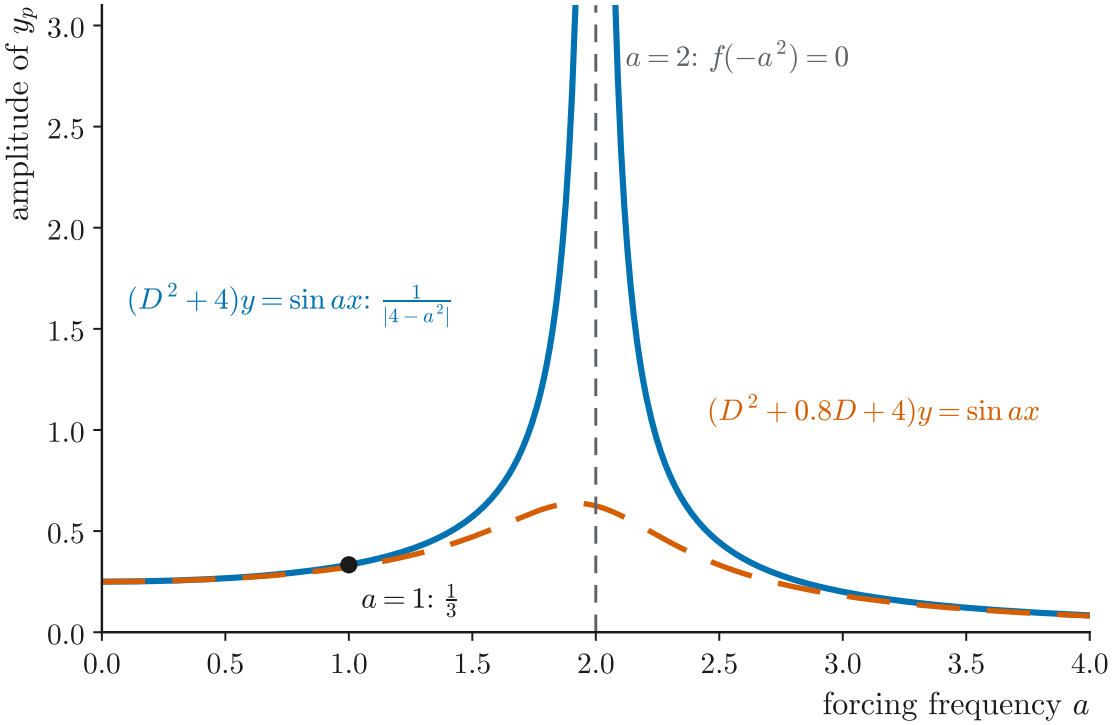


Figure G2.1. Amplitude of the PI, $\frac{1}{|f(-a^2)|}$, against the forcing frequency a. ESA ★★★

ADDED FOR CLARITY S004·4 · S004·5

How to read it. For $(D^2+4)y=\sin ax$ the PI is $\frac{\sin ax}{4-a^2}$, of amplitude $\frac{1}{|4-a^2|}$ (blue). It is $\frac{1}{4}$ at $a=0$, $\frac{1}{3}$ at $a=1$ (black dot) and blows up at $a=2$ (grey line), where $f(-a^2)=0$ and the Type 2 rule fails. With a damping term ($D^2+0.8D+4$, vermilion dashed) $f(-a^2)=4-a^2+0.8D$ never vanishes and the peak stays finite (resonance).

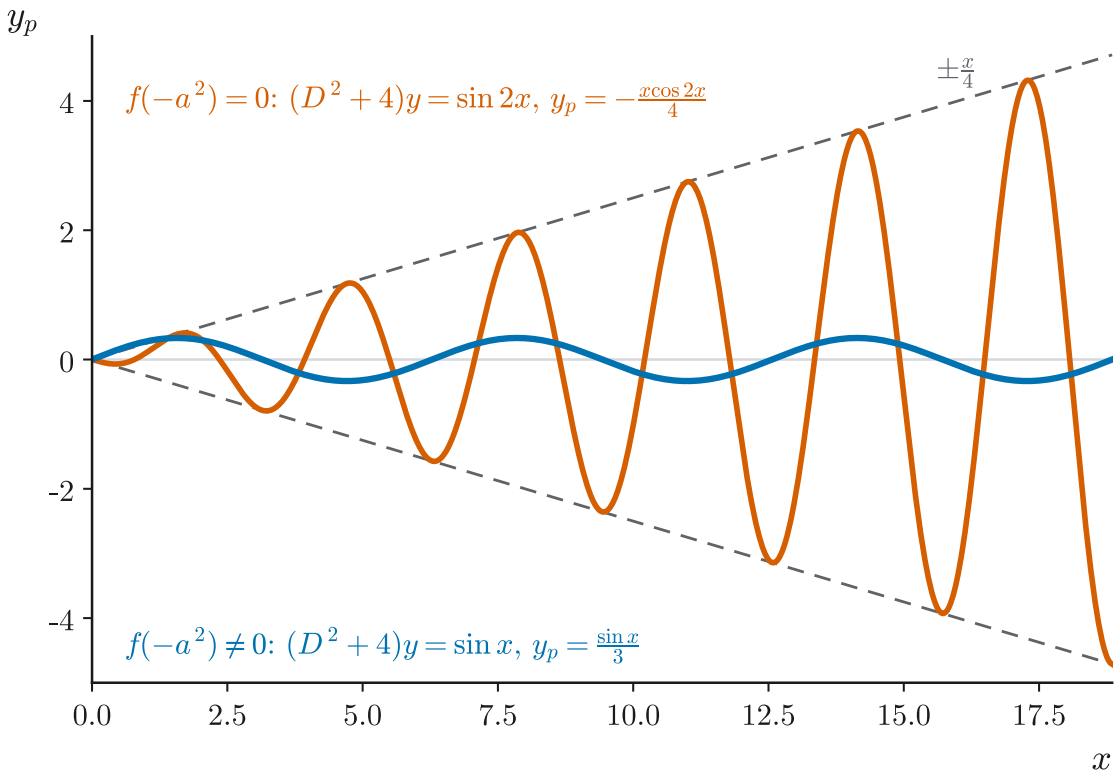
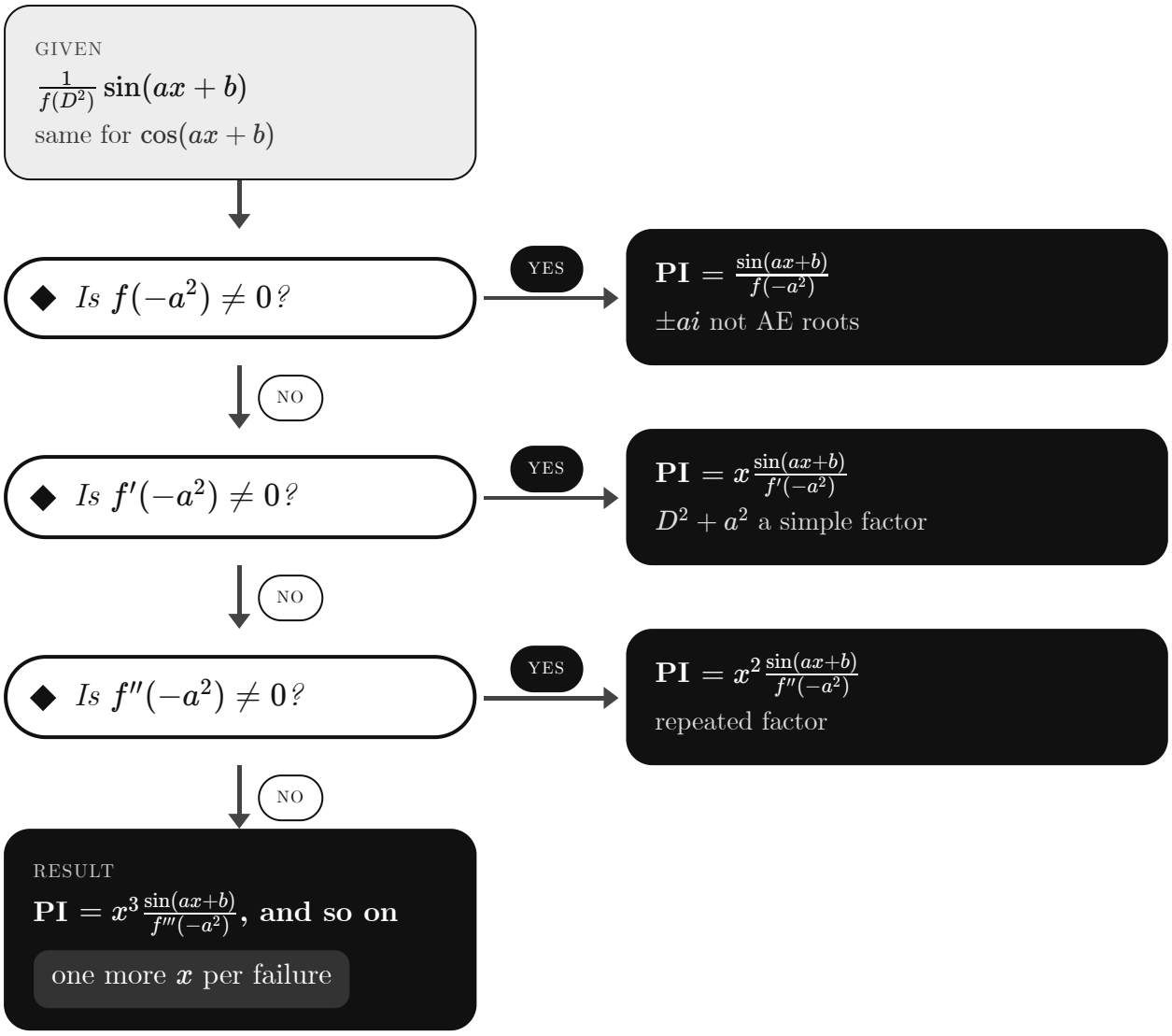


Figure G2.2. The failure case grows: $x\cos 2x$ against a bounded PI. ESA ★★★

ADDED FOR CLARITY S004·5 · S004·16

How to read it. With $(D^2+4)y=\sin 2x$ the frequency 2 is in the CF, so the PI is $-\frac{x\cos 2x}{4}$ (vermilion): an oscillation whose envelope $\pm\frac{x}{4}$ (grey dashed) grows linearly. With $\sin x$ instead ($f(-1)=3\neq 0$) the PI $\frac{\sin x}{3}$ (blue) stays bounded.

Flowcharts

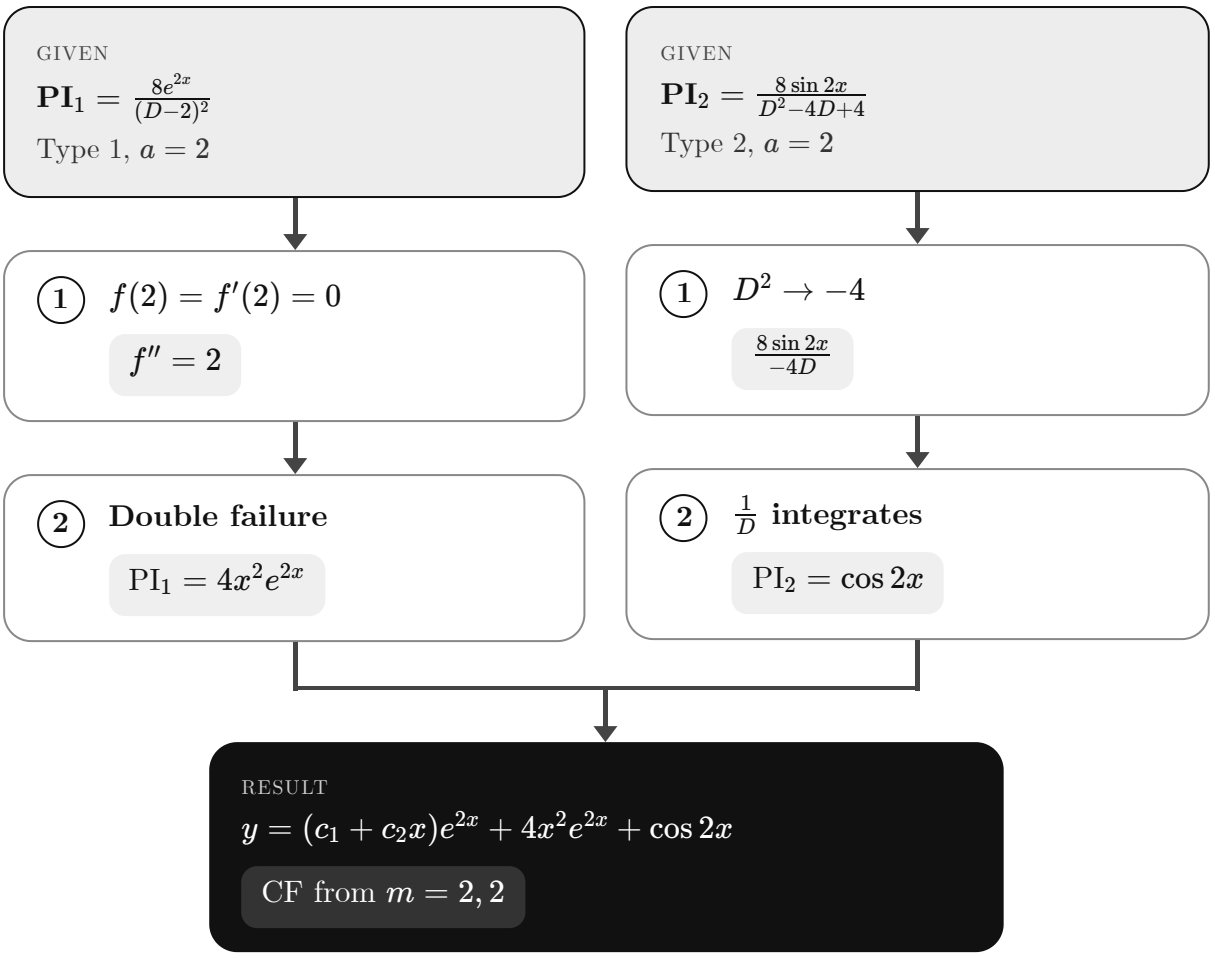


Remark. Derivatives are with respect to D; any odd D left is rationalised or, if alone, integrated ($\frac{1}{D}$).

Exam trap. Slide 004·15 writes $f'(a)$; read $f'(-a^2)$.

Figure F2.1. The Type 2 ladder ESA ★★★

S004·4 · S004·5 · S004·6 · S004·15 · T004 00:04:04 · T004 00:05:06



Remark. No rationalising is needed: the denominator is a pure D, so $\frac{1}{-4D}$ is just $-\frac{1}{4}\int$.

Figure F2.2. ESA: $(D-2)^2y = 8(e^{2x} + \sin 2x)$ ESA ★★★

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a

Important points and why

1. $f(-a^2) = 0$ exactly when $D^2 + a^2$ is a factor of $f(D)$, i.e. when $\pm ai$ are AE roots. Look at the CF first: it tells you in advance which terms will fail. ESA ★★★ T004 00:02:53 · S004·16
2. On failure, write x outside the operator and differentiate the denominator with respect to D: $\frac{\cos\sqrt{3}x}{D^2+3} = x\frac{\cos\sqrt{3}x}{2D}$. ESA ★★★ S004·16 · T004 00:23:34 · T004b 00:08:45
3. $\frac{1}{D}$ means integrate, with no constant: $\frac{x}{2D}\cos\sqrt{3}x = \frac{x}{2}\int \cos\sqrt{3}x\,dx = \frac{x\sin\sqrt{3}x}{2\sqrt{3}}$. ESA ★★★ S004·16 · T004 00:22:41 · T004b 00:09:05
4. The x stays outside: only the trigonometric function is integrated, not x times it. ESA ★★ ADDED FOR CLARITY S004·14 · T004 00:23:44
5. When only one factor of $f(D)$ fails, apply the other factor first (E3.5), then the failing one. ESA ★★ S004·13 · T004 00:21:50
6. ESA $(D-2)^2y = 8(e^{2x} + \sin 2x)$: e^{2x} fails twice (Type 1), $\sin 2x$ does not fail (Type 2 with $D^2 = -4$ leaves $-4D$). ESA ★★★ ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a

Why does an x appear when $f(-a^2)=0$?
Then $\sin ax$, $\cos ax$ solve the homogeneous equation (they are in the CF), so no multiple of them can give X. Multiplying by x gives a new function; the rule's $x\frac{1}{f'}$ is the limit of $\frac{1}{f}$ as the frequency approaches the root (same idea as Type 1).

ADDED FOR CLARITY S004·5 · T004 00:02:53

Exam trap. Slide 004·15 writes $f'(a)$ in the failure formula; it means $f'(D)$ with $D^2 \to -a^2$ (slides 004·5–6), never f' at the number a. ADDED FOR CLARITY S004·15 · S004·5

Exam trap. For $\frac{1}{D^2+a^2}\sin ax$ the answer is $-\frac{x\cos ax}{2a}$: the integral of $\sin$ brings a minus sign.
ADDED FOR CLARITY T004 00:23:44

Worked examples

W2.1 · $(D^2+3)y=\cos\sqrt{3}x$ (slide 004·16, CW P14) — ESA ★★★

GIVEN Solve $(D^2+3)y=\cos\sqrt{3}x$ (the slide asks for the PI only).

STEP 1 AE $m^2+3=0$: $m=\pm\sqrt{3}i$. CF $=c_1\cos\sqrt{3}x+c_2\sin\sqrt{3}x$.

STEP 2 $a=\sqrt{3}$, $D^2\to-3$: $f(-3)=0$, a failure.

STEP 3 E2.1: PI $=x\frac{\cos\sqrt{3}x}{2D}=\frac{x}{2}\int\cos\sqrt{3}x\,dx$.

ANSWER
$$\boxed{y=c_1\cos\sqrt{3}x+c_2\sin\sqrt{3}x+\frac{x\sin\sqrt{3}x}{2\sqrt{3}}}$$

CHECK Slide and CW answer agree; E2.2 with $a=\sqrt{3}$; sympy dsolve gives $\frac{\sqrt{3}}{6}x\sin\sqrt{3}x$, the same.

S004·16 · T004b 00:08:13 · T004b 00:09:05 · CW U2 P14

W2.2 · Type 2 term of $(D^2+4)y=x^2+\cos 2x+2^{-x}$ (ESA) — ESA ★★

GIVEN Find $\frac{1}{D^2+4}\cos 2x$ (the full question is solved in the class 005 pack).

STEP 1 $D^2\to-4$: $f(-4)=0$, a failure.

STEP 2 PI $=x\frac{\cos 2x}{2D}=\frac{x}{2}\cdot\frac{\sin 2x}{2}$.

ANSWER
$$\boxed{\frac{1}{D^2+4}\cos 2x=\frac{x\sin 2x}{4}}$$

CHECK E2.2 with $a=2$; substituting back gives $\cos 2x$ (sympy).

ESA Jan-24 s1 2d

W2.3 · $(D-2)^2y = 8(e^{2x}+\sin 2x)$ (ESA, asked three times) ESA ★★★

GIVEN Solve $(D-2)^2y = 8(e^{2x}+\sin 2x)$, where $D = \frac{d}{dx}$.

STEP 1 AE $(m-2)^2 = 0$: $m = 2, 2$. CF $= (c_1 + c_2x)e^{2x}$.

STEP 2 $\text{PI}_1 = \frac{8e^{2x}}{(D-2)^2}$: $f(2) = 0$, $f'(D) = 2(D-2)$, $f'(2) = 0$, $f'' = 2$: $\text{PI}_1 = 8 \cdot \frac{x^2e^{2x}}{2} = 4x^2e^{2x}$ (Type 1, class 003).

STEP 3 $\text{PI}_2 = \frac{8\sin 2x}{D^2-4D+4}$; $D^2 \to -4$: $\frac{8\sin 2x}{-4D} = -2 \cdot \frac{1}{D}\sin 2x$ (no failure, the lone D is integration).

STEP 4 $-2\int \sin 2x\,dx = -2 \cdot \left(-\frac{\cos 2x}{2}\right) = \cos 2x$.

ANSWER $\boxed{y = (c_1 + c_2x)e^{2x} + 4x^2e^{2x} + \cos 2x}$

CHECK sympy dsolve and checkodesol agree. CW P21 (the same with $+x^2$) prints the same terms plus $2x^2 + 4x + 3$.

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · CW U2 P21

Past-paper questions

ESA Dec 2023 · 7 marks · UE22MA141A Q3.a (= May 2024 Q3.b, June 2024 Q2.a) · solved in W2.3

Solve $(D-2)^2y = 8(e^{2x}+\sin 2x)$, where $D = \frac{d}{dx}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q2.d · solved in W2.2

Solve $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$. Type 2 part (failure): $\frac{x\sin 2x}{4}$. Full solution: class 005 pack.

SHORT-NOTE RECAP

Failure: $f(-a^2) = 0 \iff D^2 + a^2$ divides $f(D)$ ($\pm ai$ in the CF). Then PI $= x\frac{X}{f'(-a^2)}$, f' w.r.t. D; if f' also fails, $x^2\frac{X}{f''(-a^2)}$. Standard: $\frac{\sin ax}{D^2+a^2} = -\frac{x\cos ax}{2a}$, $\frac{\cos ax}{D^2+a^2} = \frac{x\sin ax}{2a}$; $\frac{\cos\sqrt{3}x}{D^2+3} = \frac{x\sin\sqrt{3}x}{2\sqrt{3}}$. ESA ×3: $(D-2)^2y = 8(e^{2x}+\sin 2x)$: $y = (c_1 + c_2x)e^{2x} + 4x^2e^{2x} + \cos 2x$.

3 Products, powers and sums: reducing to Type 2

The same rule gives the PI of $\sin ax \cos bx$, $\cos ax \cos bx$, $\sin ax \sin bx$, $\sin^2 ax$, $\cos^2 ax$, $\sin^3 ax$ and $\cos^3 ax$, after a trigonometric identity turns the product or power into a sum of single sines and cosines (and perhaps a constant, which is Type 1 with $a = 0$). Since the operator is linear, the PI of a sum is the sum of the PIs: split X, find PI_1, PI_2, ... separately and add them.

Key terms and definitions

Transformation formula. A product-to-sum identity such as $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$. S004·7 · T004 00:05:42

Superposition. $\frac{1}{f(D)}(X_1 + X_2) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2$, so $\text{PI} = \text{PI}_1 + \text{PI}_2$. S004·11 · T004 00:17:28

> *"The same formula can be used to find the PI for the following functions also. When* $X = \sin ax.\cos bx$ *or* $\cos ax.\cos bx$ *or* $\sin ax.\sin bx$*; When* $X = \sin^2 ax$ *or* $\cos^2 ax$*; When* $X = \sin^3 ax$ *or* $\cos^3 ax$*."* S004·7

Formulas

E3.1 · Product to sum ESA ★

$$\sin A \cos B = \tfrac{1}{2}[\sin(A+B) + \sin(A-B)]$$
$$\cos A \cos B = \tfrac{1}{2}[\cos(A+B) + \cos(A-B)]$$
$$\sin A \sin B = \tfrac{1}{2}[\cos(A-B) - \cos(A+B)]$$

gives	Single sines and cosines.
applies	$X = \sin ax \cos bx$, $\cos ax \cos bx$, $\sin ax \sin bx$.
note	Only the first identity is spoken in the lecture; the other two are standard.

S004·7 · T004 00:05:42

E3.2 · Squares ESA ★★★

$$\sin^2 ax = \frac{1 - \cos 2ax}{2}, \qquad \cos^2 ax = \frac{1 + \cos 2ax}{2}$$

gives	A constant (Type 1 with $a = 0$) plus a Type 2 term at frequency $2a$.
applies	$X = \sin^2 ax$ or $\cos^2 ax$.
note	$\cos^2 \frac{x}{2} = \frac{1+\cos x}{2}$, $\sin^2 \frac{x}{2} = \frac{1-\cos x}{2}$.

S004·7 · S004·13 · S004·17 · T004 00:06:23 · T004 00:20:36 · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

E3.3 · Cubes ESA ★

$$\sin^3 A = \frac{3\sin A - \sin 3A}{4}, \qquad \cos^3 A = \frac{3\cos A + \cos 3A}{4}$$

GIVES Two Type 2 terms, at frequencies a and $3a$.

APPLIES $X = \sin^3 ax$ or $\cos^3 ax$.

NOTE Listed on slide 004·7 without the identity.

ADDED FOR CLARITY S004·7 · T004 00:06:41

E3.4 · Superposition ESA ★★★

$$\frac{1}{f(D)}(X_1 + X_2 + \dots) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 + \dots = \mathrm{PI}_1 + \mathrm{PI}_2 + \dots$$

GIVES The PI of a sum of terms of different types.

APPLIES X is a sum, e.g. $e^x + \cos^2 \frac{x}{2}$.

NOTE Each term chooses its own rule (Type 1, Type 2, failure).

S004·11 · S004·14 · T004 00:17:10 · ESA Dec-23 3a

E3.5 · Factor by factor ESA ★★

$$\frac{1}{\varphi(D)\psi(D)}X = \frac{1}{\psi(D)}\left[\frac{1}{\varphi(D)}X\right]$$

GIVES A way round a failure in one factor.

APPLIES Apply the non-failing factor first, then the other.

NOTE Slide 004·13: $\frac{1}{(D-1)^2}$ on $\cos x$ first, then $\frac{1}{D^2+1}$.

S004·12 · S004·13 · T004 00:18:17 · T004 00:21:50

Graphs and diagrams

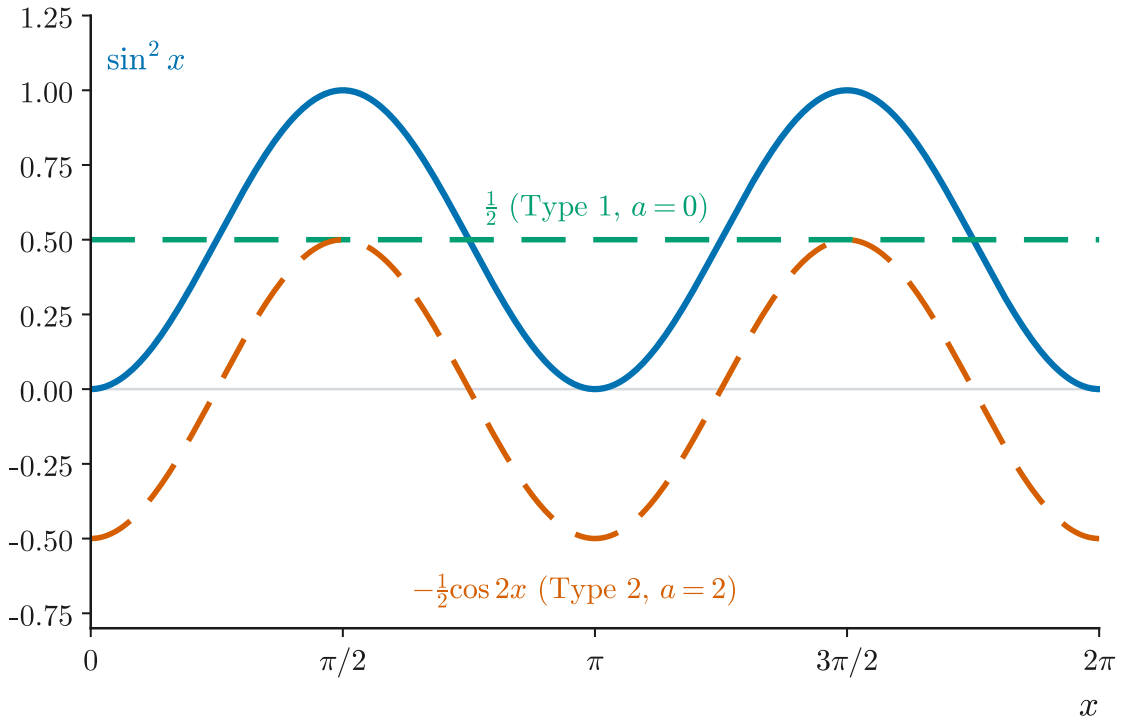
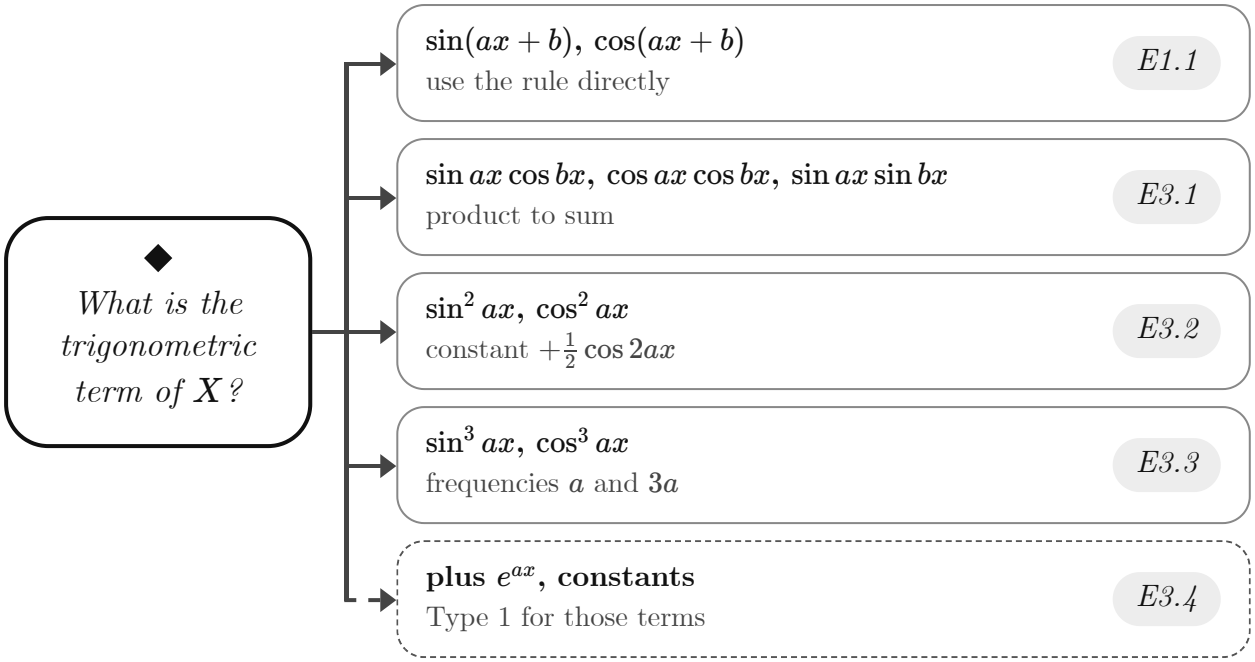


Figure G3.1. $\sin^2 x$ is a constant plus a cosine of double frequency. ESA ★★

ADDED FOR CLARITY S004·7 · S004·17

How to read it. $\sin^2 x$ (blue) never goes negative and has period π. It is the constant $\frac{1}{2}$ (green dashed, a Type 1 term with $a = 0$) plus $-\frac{1}{2}\cos 2x$ (vermilion dashed, a Type 2 term with $a = 2$, not 1). Each part gets its own rule; the two PIs are added.

Flowcharts



Remark. After rewriting, every term is either Type 1 ($D \to a$) or Type 2 ($D^2 \to -a^2$); add the PIs.

Figure F3.1. Reduce X to Type 2 first ESA ★★★

S004·7 · T004 00:05:24 · T004 00:06:12 · T004 00:20:36

Identities that reduce X to Type 2

X	Rewrite as	Rule for each term
$\sin ax \cos bx$	$\frac{1}{2}[\sin(a+b)x + \sin(a-b)x]$	Type 2 at $a+b$ and $a-b$
$\cos ax \cos bx$	$\frac{1}{2}[\cos(a+b)x + \cos(a-b)x]$	Type 2 at $a+b$ and $a-b$
$\sin ax \sin bx$	$\frac{1}{2}[\cos(a-b)x - \cos(a+b)x]$	Type 2 at $a-b$ and $a+b$
$\sin^2 ax$	$\frac{1}{2} - \frac{1}{2}\cos 2ax$	Type 1 ($D \to 0$) + Type 2 at $2a$
$\cos^2 ax$	$\frac{1}{2} + \frac{1}{2}\cos 2ax$	Type 1 ($D \to 0$) + Type 2 at $2a$
$\sin^3 ax$	$\frac{3}{4}\sin ax - \frac{1}{4}\sin 3ax$	Type 2 at a and $3a$
$\cos^3 ax$	$\frac{3}{4}\cos ax + \frac{1}{4}\cos 3ax$	Type 2 at a and $3a$

ADDED FOR CLARITY S004·7 · T004 00:05:42 · T004 00:06:23

The Type 2 term inside every ESA question

Paper, Q	$f(D)$	Type 2 term	Its PI	Full solution
Dec-23 3a (= May-24 3b, Jun-24 2a)	$(D-2)^2$	$8\sin 2x$	$\cos 2x$ (W2.3)	here, W2.3
Jul-23 3b	D^2-5D+6	$\sin x$	$\frac{\sin x+\cos x}{10}$ (W1.3)	class 005
Jan-24 s1 2d	D^2+4	$\cos 2x$	$\frac{x\sin 2x}{4}$, failure (W2.2)	class 005
Jan-24 s3 2c	D^3-2D^2+D	$\sin^2 x$	$\frac{x}{2}-\frac{\cos 2x}{25}+\frac{3\sin 2x}{100}$ (W3.2)	class 006
Jan-24 s4 2c	$(D-1)^2(D+1)^2$	$\sin^2\frac{x}{2}$	$\frac{1}{2}-\frac{\cos x}{8}$ (W3.5)	class 005

ADDED FOR CLARITY

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

Important points and why

1. No formula exists for $\cos^2\frac{x}{2}$ itself: rewrite it with an identity first, then use the rule. ESA ★★★ T004 00:20:16 · T004 00:20:36
2. The constant from $\sin^2$ or $\cos^2$ is $\frac{1}{2}e^{0x}$: Type 1 with $D\to 0$, and it fails if D is a factor of $f(D)$ ($\frac{1}{2(D^3-2D^2+D)}\to\frac{x}{2}$). ESA ★★★ S004·13 · S004·17 · T004 00:21:33 · T004b 00:09:42
3. Split a mixed right-hand side into PI_1, PI_2, ... and find each separately; the PI is their sum. ESA ★★★ S004·11 · T004 00:17:28
4. If the question says "find the PI", stop at the PI; if it says "solve", write $y=\mathrm{CF}+\mathrm{PI}$. ESA ★★ T004 00:24:12

Exam trap. Slide 004·14 (and the lecture) give the constant part of PI_2 for $(D-1)^2(D^2+1)y=\cos^2\frac{x}{2}$ as $\frac{1}{4}$; it is $\frac{1}{2}\cdot\frac{1}{f(0)}=\frac{1}{2}$. ADDED FOR CLARITY S004·14 · T004 00:21:33

Exam trap. Slide 004·17's boxed answer drops the 2 of $\frac{\cos 2x}{2(\dots)}$: the correct PI is $\frac{x}{2}-\frac{\cos 2x}{25}+\frac{3\sin 2x}{100}$ (as in CW P18), not $\frac{x}{2}-\frac{8\cos 2x}{100}+\frac{6\sin 2x}{100}$. ADDED FOR CLARITY S004·17 · T004b 00:10:35 · CW U2 P18

Exam trap. $\sin^2 x=\frac{1-\cos 2x}{2}$ (frequency doubles), not $\frac{1-\cos x}{2}$ as said at T004b 00:09:26. ADDED FOR CLARITY S004·17 · T004b 00:09:26

Worked examples

W3.1 · $(D-1)^2(D^2+1)y = e^x + \cos^2\frac{x}{2}$ (slides 004·10–14) ESA ★★★

GIVEN Solve $(D-1)^2(D^2+1)y = e^x + \cos^2\frac{x}{2}$.

STEP 1 AE $(m-1)^2(m^2+1)=0$: $m = 1, 1, \pm i$. CF $= (c_1 + c_2x)e^x + c_3\cos x + c_4\sin x$.

STEP 2 $\text{PI}_1 = \frac{1}{(D-1)^2}\left[\frac{e^x}{D^2+1}\right] = \frac{1}{(D-1)^2}\frac{e^x}{2} = \frac{x^2e^x}{2\cdot 2} = \frac{x^2e^x}{4}$.

STEP 3 PI_2: $\cos^2\frac{x}{2} = \frac{1+\cos x}{2}$. Constant: $\frac{1}{2(0-1)^2(0+1)} = \frac{1}{2}$ (slide 004·14 prints $\frac{1}{4}$).

STEP 4 Cosine: $\frac{1}{2}\cdot\frac{1}{D^2+1}\left[\frac{\cos x}{D^2-2D+1}\right] = \frac{1}{2}\cdot\frac{1}{D^2+1}\left[\frac{\cos x}{-2D}\right] = -\frac{1}{4}\cdot\frac{\sin x}{D^2+1}$.

STEP 5 Failure ($D^2+1\to 0$): $-\frac{1}{4}\cdot x\frac{\sin x}{2D} = -\frac{1}{4}\cdot x\cdot\frac{-\cos x}{2} = \frac{x\cos x}{8}$.

ANSWER
$$\boxed{y = (c_1 + c_2x)e^x + c_3\cos x + c_4\sin x + \frac{x^2e^x}{4} + \frac{1}{2} + \frac{x\cos x}{8}}$$

CHECK sympy dsolve and checkodesol agree. With the slide's $\frac{1}{4}$, substituting back leaves $-\frac{1}{4}$.

S004·10 · S004·11 · S004·12 · S004·13 · S004·14 · T004 00:14:16 · T004 00:24:01

W3.2 · $(D^3-2D^2+D)y = \sin^2 x$ (slide 004·17; ESA Jan-24 s3 2c part) ESA ★★★

GIVEN Find the PI of $(D^3-2D^2+D)y = \sin^2 x$.

STEP 1 $\sin^2 x = \frac{1-\cos 2x}{2}$, so PI $= \frac{e^{0x}}{2(D^3-2D^2+D)} - \frac{\cos 2x}{2(D^3-2D^2+D)}$.

STEP 2 Constant: $f(0)=0$, $f'(D) = 3D^2-4D+1$, $f'(0)=1$: $\frac{1}{2}\cdot\frac{x}{1} = \frac{x}{2}$.

STEP 3 Cosine: $D^2\to -4$: $-4D+8+D = 8-3D$; rationalise: $\frac{(8+3D)\cos 2x}{64-9D^2} = \frac{8\cos 2x - 6\sin 2x}{100}$.

STEP 4 Keep the factor $\frac{1}{2}$: $-\frac{1}{2}\cdot\frac{8\cos 2x - 6\sin 2x}{100}$.

ANSWER
$$\boxed{\text{PI} = \frac{x}{2} - \frac{8\cos 2x - 6\sin 2x}{200} = \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}}$$

CHECK Agrees with CW P18; the slide's boxed $\frac{x}{2} - \frac{8\cos 2x}{100} + \frac{6\sin 2x}{100}$ leaves $-\frac{\cos 2x}{2}$ when substituted (sympy).

S004·17 · T004b 00:09:26 · T004b 00:12:08 · ESA Jan-24 s3 2c · CW U2 P18

W3.3 · A product: $(D^2+1)y = \sin 2x\cos x$ ESA ★

GIVEN Solve $(D^2+1)y = \sin 2x\cos x$ (slide 004·7 lists products but works none).

STEP 1 CF $= c_1\cos x + c_2\sin x$. $\sin 2x\cos x = \frac{1}{2}(\sin 3x + \sin x)$.

STEP 2 $\frac{1}{2}\cdot\frac{\sin 3x}{D^2+1} = \frac{1}{2}\cdot\frac{\sin 3x}{-9+1} = -\frac{\sin 3x}{16}$.

STEP 3 $\frac{1}{2}\cdot\frac{\sin x}{D^2+1}$ fails: $\frac{1}{2}\left(-\frac{x\cos x}{2}\right) = -\frac{x\cos x}{4}$ (E2.2).

ANSWER
$$\boxed{y = c_1\cos x + c_2\sin x - \frac{\sin 3x}{16} - \frac{x\cos x}{4}}$$

CHECK Substituting back gives $\sin 2x\cos x$ (sympy).

ADDED FOR CLARITY S004·7 · T004 00:05:42

W3.4 · A cube: $(D^2+4)y = \sin^3 x$ ESA ★

GIVEN Solve $(D^2+4)y = \sin^3 x$.

STEP 1 CF $= c_1\cos 2x + c_2 \sin 2x$. $\sin^3 x = \frac{3\sin x - \sin 3x}{4}$ (E3.3).

STEP 2 $\frac{3}{4}\cdot\frac{\sin x}{-1+4} = \frac{\sin x}{4}$; $-\frac{1}{4}\cdot\frac{\sin 3x}{-9+4} = \frac{\sin 3x}{20}$.

ANSWER $$\boxed{y = c_1\cos 2x + c_2\sin 2x + \frac{\sin x}{4} + \frac{\sin 3x}{20}}$$

CHECK Substituting back gives $\sin^3 x$ (sympy).

ADDED FOR CLARITY S004·7 · T004 00:06:41

W3.5 · Type 2 term of $(D-1)^2(D+1)^2 y = \sin^2\frac{x}{2} + 2^{-x} + x$ (ESA) ESA ★★

GIVEN Find $\frac{1}{(D-1)^2(D+1)^2}\sin^2\frac{x}{2}$ (the full question is solved in the class 005 pack).

STEP 1 $(D-1)^2(D+1)^2 = (D^2-1)^2$; $\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$.

STEP 2 Constant: $\frac{1}{2}\cdot\frac{1}{(0-1)^2} = \frac{1}{2}$. Cosine: $D^2 \to -1$: $(-1-1)^2 = 4$, so $-\frac{1}{2}\cdot\frac{\cos x}{4}$.

ANSWER $$\boxed{\frac{1}{(D^2-1)^2}\sin^2\frac{x}{2} = \frac{1}{2} - \frac{\cos x}{8}}$$

CHECK Agrees with CW P22; substituting back gives $\sin^2\frac{x}{2}$ (sympy).

ESA Jan-24 s4 2c · CW U2 P22

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q2.c (= CW P18) · solved in W3.2

Solve $(D^3-2D^2+D)y = x^2e^{-x} + \sin^2 x$. $\sin^2 x$ part: $\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$. Full solution: class 006 pack.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (= CW P22) · solved in W3.5

Solve $[(D-1)^2(D+1)^2]y = \sin^2\frac{x}{2} + 2^{-x} + x$. $\sin^2\frac{x}{2}$ part: $\frac{1}{2} - \frac{\cos x}{8}$. Full solution: class 005 pack.

SHORT-NOTE RECAP

Reduce first: $\sin A\cos B = \frac{1}{2}[\sin(A+B)+\sin(A-B)]$; $\sin^2 = \frac{1-\cos 2ax}{2}$, $\cos^2 = \frac{1+\cos 2ax}{2}$; $\sin^3 A = \frac{3\sin A - \sin 3A}{4}$. The constant is Type 1 ($D \to 0$; fails if D divides f). Add the separate PIs. $(D-1)^2(D^2+1)y = e^x + \cos^2\frac{x}{2}$: PI $= \frac{x^2e^x}{4} + \frac{1}{2} + \frac{x\cos x}{8}$ (slide prints $\frac{1}{4}$). $\frac{\sin^2 x}{D^3-2D^2+D} = \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$ (slide drops a 2).

A Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.2, E2.1
$f(D)$, $f(D^2)$	Operator polynomial	$f(D^2)$: $f(D)$ written through D^2 where possible	E1.1
$f(-a^2)$	Replaced denominator	Every D^2 replaced by $-a^2$ (minus a squared)	E1.1, E2.1
$f'(D)$, $f''(D)$	Derivatives w.r.t. D	Used in the failure cases; slide 004·15 writes $f'(a)$	E2.1, E2.3
a	Frequency	In $\sin(ax+b)$; not the exponent of Type 1 (same letter)	E1.1
b	Phase constant	Never enters the rule	E1.1
$p+qD$	Linear remainder	What is left after $D^2 \to -a^2$; rationalised with $p-qD$	E1.3
PI_1, PI_2	Partial PIs	PIs of the separate terms of X	E3.4
X	Right-hand side	A function of x; capital X, small x	E1.1

B Sources, coverage and gaps

Source	Path	Use
Deck 004 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/004_Particular_integral_of_Type_2__03746afc7474a254.*	18 slides; content on 3–17 (every equation read from the rendered images: the .slides.md has none)
Transcript T004 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.*	24.7 min; slides 4–14
Transcript T004b (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.*	12.6 min; 00:06:12–00:12:28 Type 2 constraints and slides 16–17 (00:00–06:04 is Type 1, used by the class 003 pack)
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problems 13 and 14 (solved in full); Type 2 terms of problems 18, 21 and 22
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Dec-23 3a (= May-24 3b = Jun-24 2a) in full; Type 2 parts of 4 more questions, via the shared ESA map
Videos 004 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_of_standard_functions_-_Type_2__d25d020162de0016.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences; 7 distinct / 9 occurrences are PI by operator rules). Type 2 sits in 5 distinct questions (7 occurrences): $(D-2)^2y = 8(e^{2x} + \sin 2x)$ in Dec-23 3a = May-24 3b = Jun-24 2a (★★★, 6–7 marks, solved in full here, W2.3), and the Type 2 term of the mixed questions Jul-23 3b ($\sin x$), Jan-24 set 1 2d ($\cos 2x$ with a failure), set 3 2c ($\sin^2 x$, = CW P18 = slide 004·17) and set 4 2c ($\sin^2 \frac{x}{2}$), whose other terms are solved in the class 005 and 006 packs (table C3.2). Stars: the rule, rationalising and the failure case ★★★; $\sin^2/\cos^2$ ★★★ (three questions); products $\sin A \cos B$ and $\sin^3$ ★ (slides only, never asked); the second failure (x^2, f'') ★ (never asked).

Known gaps, conflicts and corrections

1. Slide 004·14 (and the lecture, T004 00:21:33) gives the constant part of PI_2 for $(D-1)^2(D^2+1)y = e^x + \cos^2\frac{x}{2}$ as $\frac{1}{4}$. It is $\frac{1}{2}$: $\frac{1}{2} \cdot \frac{1}{f(0)} = \frac{1}{2} \cdot \frac{1}{(0-1)^2(0+1)} = \frac{1}{2}$. With $\frac{1}{4}$, substituting back leaves $-\frac{1}{4}$ (sympy). Correct PI: $\frac{x^2 e^x}{4} + \frac{1}{2} + \frac{x\cos x}{8}$.

2. Slide 004·17 drops the factor 2 of $\frac{\cos 2x}{2(D^3-2D^2+D)}$ in its boxed answer (T004b 00:10:35 says "just ignore this two"). The correct PI of $(D^3-2D^2+D)y=\sin^2 x$ is $\frac{x}{2}-\frac{8\cos 2x}{200}+\frac{6\sin 2x}{200}=\frac{x}{2}-\frac{\cos 2x}{25}+\frac{3\sin 2x}{100}$, which agrees with the CW P18 answer; the slide's $\frac{x}{2}-\frac{8\cos 2x}{100}+\frac{6\sin 2x}{100}$ leaves $-\frac{\cos 2x}{2}$ (sympy).
3. Slide 004·15 writes the failure results as $x\frac{\sin(ax+b)}{f'(a)}$ and $x^2\frac{\sin(ax+b)}{f''(a)}$, while slides 004·5–6 and the conditions on slide 004·15 itself use $f'(-a^2)$, $f''(-a^2)$. Read all of them as: differentiate $f(D)$ with respect to D, then replace D^2 by $-a^2$ and treat the remaining odd D as integration (slide 004·16 does exactly this: $\frac{x}{2D}\cos\sqrt{3}x$).
4. The precise standard results $\frac{1}{D^2+a^2}\sin ax=-\frac{x\cos ax}{2a}$ and $\frac{1}{D^2+a^2}\cos ax=\frac{x\sin ax}{2a}$ are added for clarity (E2.2); they follow from the slides' failure rule.
5. Slide 004·5's second line has a missing bracket, $\cos(ax+b$; slide 004·16 writes $\cos\sqrt{3}x$, meaning $\cos(\sqrt{3}x)$.
6. Slide 004·7 lists $\sin^3 ax$ and $\cos^3 ax$ but gives no identity and no example; the identities $\sin^3 A=\frac{3\sin A-\sin 3A}{4}$, $\cos^3 A=\frac{3\cos A+\cos 3A}{4}$, the product-to-sum table and the examples W3.3 and W3.4 are added for clarity.
7. T004b 00:09:26 says $\sin^2 x=\frac{1-\cos x}{2}$; the slide (004·17) and the working use $\frac{1-\cos 2x}{2}$, which is correct. T004b 00:11:48 says "64 minus 36 D^2"; it is $64-9D^2$ (= 100 after $D^2=-4$), as on the slide.
8. T004b 00:07:32 says "in case it becomes zero, we go by case one"; it means the same multiply-by-x idea as Type 1, applied with $f'(D)$ and then $D^2\to -a^2$.
9. The lecture's opening example "$D+D^2+D^3$" (T004 00:01:42) is not worked; W1.4 finishes it for $X=\sin 2x$ (added for clarity).
10. Slides 004·9 and 004·14 stop at $y=y_c+y_p$ and "GS is y = CF + PI"; the general solutions are written out here.
11. No other slide answer in this class is wrong: slides 004·8–9, 004·10–13 and 004·16 were recomputed (sympy dsolve and checkodesol). CW P13 ($\frac{3\cos 3x+\sin 3x}{50}$) and CW P14 ($\frac{x\sin\sqrt{3}x}{2\sqrt{3}}$) answers are right.
12. ESA: the full mixed questions Jul-23 3b, Jan-24 set 1 2d and set 4 2c are solved in the class 005 pack and Jan-24 set 3 2c in the class 006 pack; here only their Type 2 terms are worked (table C3.2).
13. Transcript recognition errors were corrected silently: "f of df" / "f of d" → $f(D)y$, $f(D)$; "Casi X" / "cos e x" / "car square" / "ka square" → $\cos ax$, $\cos^2$; "dq" / "mq" / "AQ" → D^3, m^3, a^3; "b square plus a square" → D^2+a^2; "psi beta x" → $\sin\beta x$; "plus r minus" → $\pm$; "60 into minus 4" → $16\times(-4)$; "Pi 1" → PI_1; "k power X" → e^{ax}; "row 3x" → $\sqrt{3}x$; "replace t square" → replace D^2; "replace d by gate" → replace D by 0; "minus 40" → $-4D$; "a T into cost 3x" → $8\cos 2x$.

Coverage

Source	Citations	Slides cited (slide × citations)
S004	82	3×1, 4×8, 5×11, 6×3, 7×10, 8×2, 9×10, 10×1, 11×4, 12×2, 13×5, 14×4, 15×6, 16×8, 17×7
CW	6	
ESA	23	
T004	60	
T004b	22	

Slides 1, 2 and 18 are title and closing slides; slide 3 is the class content. The Type 2 lecture (T004, 24.7 min) covers slides 4–14; the newer problem lecture (T004b, 12.6 min) covers slides 15–17 after a Type 1 recap (00:00–06:04, cited by the class 003 pack, where it is tagged T004a). Bucket numbers are not contiguous (the decks say SESSION 1–11; 007, 008, 010, 012 and 013 do not exist; 009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11); this deck is SESSION 4.